Adjustment-set audit: antidepressants in pregnancy and child autism

Section 7 of the simulation protocol. Draft v0.6, 27 October 2026. Vera Wilde, TUM Campus Heilbronn. Rai et al. 2017, Brown et al. 2017, Sujan et al. 2017, Suarez et al. 2022 and Chan et al. 2026 are coded from their full texts and supplements.

Coding scheme

Each variable is coded against one causal diagram with these paths: maternal genetic liability and family severity (shared across pregnancies) and pregnancy-specific severity affect both use and autism; use affects preterm birth, which affects autism; paternal liability and depression affect paternal use and autism. Following Cinelli, Forney and Pearl (2024) and Hünermund, Louw and Rönkkö (2025):

Code	Meaning	Consequence of adjusting
C	Confounder: a common cause of use and outcome, measured before exposure starts	Removes bias
Px	Proxy: a measured stand-in for a confounder, such as recorded diagnoses for severity	Removes part of the bias; residual confounding remains
Pr	Precision: affects the outcome only, such as child sex	Harmless; may tighten intervals
T	Timing problem: measured after exposure starts, so it may be partly a consequence of exposure	May remove part of a real effect, or open a biasing path
M	Mediator: on the path from drug to outcome, such as preterm birth	Removes part of the total effect
D	Design choice: a restriction, exclusion or comparison window that changes who is compared	Changes the estimand; may build in assumptions

Rai et al. 2017, BMJ (Stockholm Youth Cohort), coded from full text

Exposure: antidepressant use from 30 days before pregnancy to birth, from antenatal self-report (median interview at 10 weeks) or prescription register; either source counts. Main comparison: exposed children versus children of mothers with a psychiatric disorder who did not use antidepressants. Main estimate aOR 1.45 (1.13 to 1.85); sibling OR 1.36 (0.84 to 2.20), from 94 sibling sets discordant on exposure and outcome (appendix); paternal negative control aOR 1.13 (0.68 to 1.88). The authors concluded the association might not be due to confounding alone. No dose-response analysis (small numbers, per the authors).

Variable	Analyses	Code	Note
Birth year	Main, PS (exact), sibling, paternal	C	Secular trends in prescribing and diagnosis.
Maternal psychiatric diagnoses "before birth" (depression, anxiety, bipolar, psychoses, OCD, stress-related, other)	Main (model 2), PS, paternal	Px + T	Proxy for indication and severity. "Before birth" includes diagnoses made during pregnancy, which can follow from care contact tied to the prescription. Needs the split into pre-pregnancy and in-pregnancy diagnoses.
Number of depression diagnoses, type of care, prior care episodes	PS (some exact-matched)	Px	Severity proxies. Same timing question as above.
Other maternal medications in pregnancy (antiepileptics, antipsychotics, anxiolytics)	Main (model 2), PS	T	Co-exposures during pregnancy. Can mark severity (confounder) or follow from antidepressant treatment (augmentation, switching). Direction of bias depends on which.
Smoking and BMI at first antenatal visit	PS	C / T	Measured after exposure starts (exposure window opens 30 days before pregnancy). Mostly pre-pregnancy habits; low concern.

Variable	Analyses	Code	Note
Maternal and paternal age, parity, education, income fifth, birth country	Main (model 3), PS, paternal	C	Standard sociodemographic confounders.
Child sex	Main, PS (exact), sibling, paternal	Pr	Harmless.
Comparison group restricted to mothers with a psychiatric diagnosis	Main	D	Active-indication comparison. Reduces confounding by indication but conditions on recorded diagnosis, which partly depends on care seeking.
Maternal antidepressant use and paternal depression	Paternal negative control	C	Correct for the negative-control contrast. The control is still valid only if paternal use has no causal path to the child.
Exposure window from 30 days before pregnancy	All	D	Includes part of the preconception period in "exposed"; no separate preconception analysis for mothers.

Brown et al. 2017, JAMA (Ontario public drug coverage), coded from full text and supplement

Exposure: two or more consecutive SSRI or SNRI prescriptions between conception and delivery. Estimates: crude HR 2.16 (1.64 to 2.86); multivariable HR 1.59 (1.17 to 2.17); HDPS-weighted HR 1.61 (0.997 to 2.59); HDPS-matched HR 1.64 (1.07 to 2.53); sibling HR 1.60 (0.69 to 3.74), from 620 discordant pairs with 11 versus 8 cases. Mood and anxiety cohort, HDPS-weighted: 1.62 (1.04 to 2.52); SSRIs 1.78 (1.07 to 2.95); second or third trimester 1.94 (1.20 to 3.15). Recent users (90 to 365 days before conception only): HR 1.85 (1.37 to 2.51).

Variable or choice	Analyses	Code	Note
Gestational age at birth	HDPS weighting (prespecified covariate)	M	Affected by exposure (exposed mean 38.3 vs 38.7 weeks). Including it in the propensity score adjusts away any effect operating through shorter gestation, in the paper's main estimate.
Covariate selection rule for the multivariable model: "baseline" variables with standardized difference above 0.10	Multivariable model	T / M	A data-driven rule, not a causal one. By Table 1 the variables above 0.10 include psychiatric admissions in pregnancy (0.28), psychiatric emergency visits in pregnancy (0.36), serotonergic pain (0.30) and nausea (0.32) medication, obstetrician prenatal care (0.19) and gestational age at birth (0.18). The exact list is not printed for the main model.
Psychiatric hospitalization and emergency visits in pregnancy; other serotonergic medications	Multivariable model, mood/anxiety cohort (eTable 7)	T	Listed explicitly in the eTable 7 footnote. Measured after exposure starts.
HDPS empirical covariates: top 500 codes from the 365 days before conception	HDPS weighting	Px	Pre-exposure window, so clean on timing. Pregnancy codes in eTable 3 come from earlier pregnancies. The authors note residual imbalance on psychiatric diagnoses and severity after weighting.
Mood or anxiety disorder in the 2 years before conception; maternal age, parity, year, income, rural residence, child sex	HDPS weighting	C / Pr	Pre-exposure confounders; child sex is a precision variable.

Variable or choice	Analyses	Code	Note
Reference group excludes women with a prescription in the 90 days before conception; those 2,210 deliveries dropped entirely	All	D	Removes the preconception-exposed from both arms. The separate recent-user comparison (90 to 365 days before) is read by the authors as evidence of confounding; it overlaps oocyte maturation.
Single-prescription pregnancies excluded (1,501)	All	D	Removes brief and early exposure, including the window of greatest teratogenic concern.
One birth per woman, randomly selected	Main cohort	D	Drops later pregnancies at random; reduces clustering but not a bias concern.
Sibling model: maternal age, parity, calendar year only; closest consecutive sibling; births from 1997	Sibling	C / D	Shared factors removed by design; non-shared severity unmeasured. Exposed sibling is usually the later birth (primiparous 6.1% vs 15.7%). With 19 cases in total, the interval 0.69 to 3.74 cannot resolve the question either way. No paternity data, so half-siblings cannot be identified.

Sujan et al. 2017, JAMA (Sweden, 1.58 million offspring), coded from full text

Autism estimates: crude HR 2.02 (1.80 to 2.26); adjusted 1.64 (1.46 to 1.83); sibling 0.83 (0.62 to 1.13). Dispensing before pregnancy only: adjusted HR 1.40 (1.02 to 1.93), versus 1.75 (1.49 to 2.07) for first-trimester dispensing. Paternal first-trimester dispensing: HR 1.31 (1.05 to 1.62). Sibling sample (eTable 1): 10,975 exposed and 13,994 unexposed differentially exposed siblings, with 136 and 274 autism diagnoses.

Variable or choice	Analyses	Code	Note
Parity, year of birth	All models	C	Pre-exposure.
Maternal and paternal country of birth, age, education, criminal conviction	Population and timing models	C	Pre-exposure family characteristics.
Severe psychiatric illness (inpatient schizophrenia, bipolar, other psychoses); suicide attempts	Population and timing models	Px	The only indication proxies. Depression and anxiety, the actual indications, are not adjusted for; residual confounding by indication is likely large.
First-trimester exposure window: 90 days before to 90 days after conception	All	D	"First-trimester" exposure includes three months before conception.
"Before pregnancy only": dispensing 270 to 90 days before conception, none in pregnancy or 180 days after	Timing comparison	D	Treated as drug-free in its effect. Overlaps oocyte maturation; also excludes women who restart after delivery, a selection on later course.
Paternal model adjusts for parity and year of birth only	Paternal negative control	D	Paternal and maternal first-trimester dispensing are strongly associated (OR 3.42, 3.26 to 3.58), but maternal use is not adjusted for. The paternal HR of 1.31 therefore partly carries maternal exposure itself and cannot separate confounding from a maternal drug effect.

Variable or choice	Analyses	Code	Note
Sibling covariates: factors varying within families (parity, parental age)	Sibling	C	Shared factors removed by design; non-shared severity unmeasured.

Suarez et al. 2022, JAMA Internal Medicine (US Medicaid MAX and MarketScan), coded from full text and supplement

Exposure: at least one antidepressant dispensing from day 127 after LMP to delivery; unexposed: none from LMP to delivery. Autism, pooled (Figure 2): crude HR 1.81 (1.69 to 1.92); adjusted 1.06 (0.96 to 1.17); HDPS 1.07 (0.97 to 1.18); discontinuer referent 1.08 (0.94 to 1.24); sibling HR 0.86 (0.60 to 1.23) in the main text, 0.87 (0.60 to 1.25) in eTable 21, from 163 informative sibling sets. Early-pregnancy window (LMP to day 126, eTable 4): crude HR 1.83 (1.73 to 1.93); adjusted 1.07 (0.97 to 1.17); discontinuer referent 1.03 (0.88 to 1.20).

Variable or choice	Analyses	Code	Note
Antidepressant indications (depression, anxiety, ADHD, bipolar and others), assessed 90 days before LMP through the day before delivery	PS, HDPS, sibling	Px + T	The assessment window covers the whole exposure period. Diagnoses recorded during pregnancy can follow from the care contact that produced the prescription.
Other psychotropic medications (benzodiazepines, anxiolytics, antipsychotics, stimulants and others), same window	PS, HDPS, sibling	T	Co-medication during the exposure window: may mark severity or follow from antidepressant treatment.
Lifestyle (smoking, alcohol, substance use, poor nutrition, hyperemesis), same window	PS, HDPS, sibling	T	Partly post-exposure; nausea and vomiting in pregnancy can be treated with serotonergic drugs.
Comorbidities including the obstetric comorbidity index, infections; same window	PS, HDPS, sibling	T	The obstetric index includes pregnancy complications that can follow exposure.

Variable or choice	Analyses	Code	Note
Prenatal variables (adequacy of prenatal care, month care started, prenatal vitamins, multiple gestation), same window	PS, HDPS, sibling	T / C	Care adequacy is measured across pregnancy; month care started is pre-exposure for the late window.
Severity proxies (psychiatric visits, hospitalizations, emergency visits, number of diagnoses and psychotropics), 90 days before LMP through day 126	PS, HDPS, sibling	Px	Ends the day before the late exposure window opens, so clean on timing for the main analysis; not for the early-window analysis.
200 empirically selected HDPS covariates	HDPS	Needs window	Assessment window for the empirical covariates not stated in eTable 2.
Sibling model adjusts for all PS covariates except race, state and county socioeconomic level, plus birth order	Sibling	T	Carries the same post-exposure covariates into the sibling comparison.
Discontinuers: dispensing 90 to 31 days before LMP, none from 30 days before LMP to delivery	Discontinuer referent	D	Treated as unexposed in effect. Overlaps oocyte maturation, so the comparison assumes no preconception pathway.
Exposure classes pooled (SSRIs, SNRIs, tricyclics, bupropion) in the main estimate	Main	D	Class- and drug-specific results are in eTables 5 to 12; sibling analyses were not run by class.

Variable or choice	Analyses	Code	Note
Interpretive claim: sibling comparisons "provide an estimate of the direct causal effect"	Discussion	Claim	A sibling comparison removes shared confounding; it does not by design turn a total-effect estimate into a direct effect. The same paragraph concedes sibling estimates are more vulnerable to exposure misclassification, biasing toward the null.

Chan et al. 2026, Lancet Psychiatry (meta-analysis, 37 studies), coded from full text and appendix

Autism, pooled RR (Figure 3): any antidepressant in pregnancy 1.69 (1.24 to 2.30), k=25; restricted to mothers with mental disorders 1.32 (1.16 to 1.50), k=10; discontinuers as reference 1.14 (1.00 to 1.31), k=5; sibling-matched 0.98 (0.86 to 1.12), k=7; paternal use during pregnancy 1.28 (1.16 to 1.40), k=6; paternal use 3 months before or during conception 1.13 (0.97 to 1.33), k=3; low dose 1.23 (1.03 to 1.45), high dose 1.78 (1.42 to 2.21), k=2 each.

Reading in the paper	Estimate	Code	Note
"No evident dose-response relationship, arguing against a causal role"	High dose 1.78 (1.42 to 2.21) vs low dose 1.23 (1.03 to 1.45)	Claim	The two intervals overlap. The conclusion rests on a between-group test that could not separate two estimates from two studies each, and a comparison that cannot separate the groups is not evidence against a dose-response.
"Paternal antidepressant use around conception was not linked to ASD"	1.13 (0.97 to 1.33), k=3	Claim	Compatible with no association and with up to a 33% increase. The interval overlaps the during-pregnancy estimate, so the two windows cannot be told apart; it neither rules out a sperm pathway nor favors a familial reading.
Paternal use during pregnancy as a negative control	1.28 (1.16 to 1.40), k=6	D	Pooled from studies such as Sujan that do not adjust the paternal model for maternal use; paternal and maternal use are strongly associated (Sujan OR 3.42).
Sibling-matched pooled estimate	0.98 (0.86 to 1.12), k=7	D	Pools sibling estimates that each carry the design's known biases (non-shared confounding, amplified misclassification, half-siblings); pooling adds precision, not validity.
Analyses that keep a signal	Mental-disorder restricted 1.32 (1.16 to 1.50); discontinuer reference 1.14 (1.00 to 1.31)	Note	Both active-comparator designs remain above 1 for autism; the abstract summarizes all sensitivity analyses as attenuated or non-significant.

Cross-study flags so far

- **Post-exposure adjustment is routine.** Suarez measures nearly all propensity-score covariates from 90 days before LMP to the day before delivery, spanning the exposure window, and carries them into the sibling model.
- **Mediator adjustment.** Brown's main HDPS weighting includes gestational age at birth, and its multivariable models select covariates by standardized difference, which pulls in psychiatric hospitalization and emergency visits in pregnancy and serotonergic co-medication. Rai adjusts for psychiatric diagnoses and co-medications recorded up to birth. Any diagnosis or prescription that follows from antidepressant treatment is a post-exposure variable.
- **Comparison windows build in the conclusion.** Sujan's before-pregnancy group and Suarez's discontinuer group (90 to 30 days before the last menstrual period) are treated as drug-free in their effect. Both windows overlap oocyte maturation, so each comparison assumes there is no preconception pathway.
- **Exposure definitions remove early exposure.** Brown excludes single-prescription pregnancies; Suarez's primary window starts at gestational week 19. Neither design looks squarely at the first trimester.
- **Sujan's paternal control is not adjusted for maternal use.** With paternal and maternal dispensing strongly associated (OR 3.42), a paternal HR of 1.31 is what a maternal drug effect would also produce.
- **The paternal negative control is used both ways.** Rai reports a paternal aOR of 1.13 (0.68 to 1.88), an interval too wide to resolve the question either way, and still reads the maternal association as possibly causal. Sujan and Chan read a paternal association of similar size as evidence of confounding.
- **Sibling intervals are wide.** Rai 0.84 to 2.20; Brown 0.69 to 3.74 (620 pairs); Sujan 0.62 to 1.13. Both cover a substantial increase. The sibling comparisons did not separate the arms and could not have, at that size.